INCH-POUND

MIL-DTL-6363/8B
6 May 2024
SUPERSEDING
MIL-L-6363/8A
30 September 1997

DETAIL SPECIFICATION SHEET

LAMPS, INCANDESCENT, AIRCRAFT SERVICE, SINGLE CONTACT
MIDGET FLANGED BASE T-1-3/4 BULB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-6363.

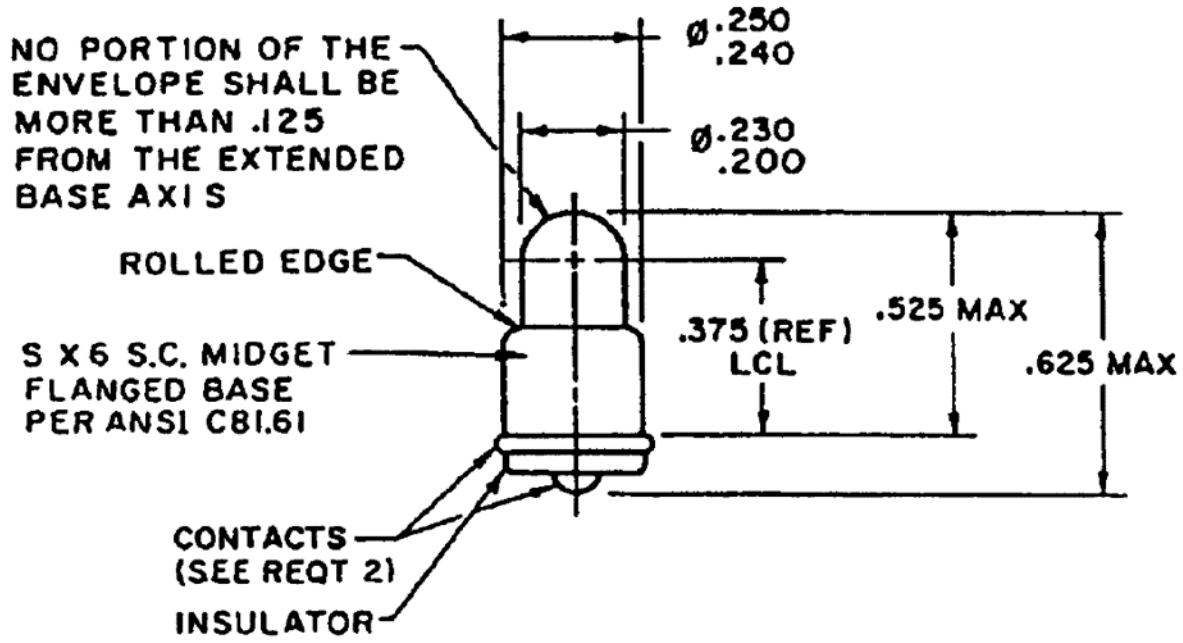


FIGURE 1. Configuration and dimensions.

See NOTES on the next page.

AMSC N/A FSC 6240

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NOTES:

1/ Dimensions are in inches.

2/ Intended use – aircraft indicator, instrument, and panel light assemblies.

3/ For design feature purposes, this specification sheet takes precedence over procurement documents.

4/ Qualification – Lamp M6363/8-7AS15 (or optional M6363/8-5 AS15) shall be fully qualified in accordance with table I of MIL-DTL-6363. Lamp M6363/8-2 AS10 shall be qualified in accordance with table II of MIL-DTL-6363. Red lamp M6363/8-7R (or optional -5R) shall be qualified in accordance with table II of MIL-DTL-6363.

5/ Qualification by similarity – All other basic lamps (Mean Spherical Candela [MSCD] ±25 percent) and "AS" designated lamps may be granted qualification by similarity and extension of qualification from the qualified lamps with no additional testing, as approved by NAVSEA. All other colored lamps may be granted qualification from -7R (or option -5R) with no additional testing, as approved by NAVSEA.

FIGURE 1. Configuration and dimensions. – Continued

REQUIREMENTS:

1. Configuration and dimensions: See figure 1.

2. Construction:

   a. Standard lamp construction shall include a side weld or solder bead and solder bead bottom contact.

   b. Where required for greater reliability and added resistance to corrosion and internal shorting, solderless construction is available as an option by the addition of a dash and suffix letter "S" to the identification number in table I.

EXAMPLE: M6363/8-1-S

When so specified, solderless construction shall include a side weld and nickel-plated brass bottom contact.

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TABLE I. Part or identifying numbers (PIN) and operating characteristics.

Part or Identifying Number (PIN) (See REQT 2)	Type MIL-DTL-6363	Bulb Finish	Shock Test Level (G's)	Electrical Ratings			Filament Style	MSCD 1/ 2/	Average Rated Lab Life at DC
				Volts (Nom)	Amperes (±10%)	Watts (Max)		Candelas	Hours
M6363/8-1	I	Clear	50	5.0	0.06	0.33	C-2R	0.15±25%	2,000
M6363/8-1 AS15		Clear		5.0	0.06	0.33	C-2R	0.15±15%	2,000
M6363/8-2		Clear		6.0	0.20	1.32	C-2R	0.34±25%	1,000
M6363/8-2 AS10		Clear		6.0	0.20	1.32	C-2R	0.34±10%	1,000
M6363/8-2R		Red		6.0	0.20	1.32	C-2R	0.34±25%	1,000
M6363/8-2B		3/		6.0	0.20	1.32	C-2R	0.34±25%	1,000
M6363/8-3		Clear		6.3	0.20	1.38	C-2F	0.40±25%	10,000
M6363/8-3 AS15		Clear		6.3	0.20	1.38	C-2F	0.40±15%	10,000
M6363/8-4		Clear		14.0	0.10	1.54	C-2F	0.50±25%	5,000
M6363/8-4 AS15		Clear		14.0	0.10	1.54	C-2F	0.50±15%	5,000
M6363/8-5		Clear		28.0	0.04	1.24	C-2F	0.34±25%	1,500
M6363/8-5 AS15		Clear		28.0	0.04	1.24	C-2F	0.34±15%	1,500
M6363/8-5R		Red		28.0	0.04	1.24	C-2F	0.34±25%	1,500
M6363/8-6 AS15		Clear		28.0	0.04	1.24	C-2F	0.15±15%	5,000
M6363/8-7		Clear		28.0	0.04	1.24	C-2F	0.30±25%	2,000
M6363/8-7 AS15		Clear		28.0	0.04	1.24	C-2F	0.30±15%	2,000
M6363/8-7R		Red		28.0	0.04	1.24	C-2F	0.30±25%	2,000

NOTES:

1/ MSCD for -2 and -2 AS10 is when operated at 5 volts.

2/ MSCD for colored lamps applies to clear lamp before coating is applied.

3/ Blue/white

3. Finish: The bulb's glass globe finish shall be as indicated in table I. Red lamps shall conform to SAE AS25050 for instrument and panel lighting red. Blue/white lamps shall conform to SAE AS25050 for blue filtered white color.

4. Lamp operating characteristics: When operating at rated voltage, the lamp operating characteristics shall be as specified in table I.

5. Where indicated by suffix "AS_" in the part number, lamps shall be aged for a minimum of 10 hours at rated voltage and selected for 15 or 10 percent tolerance on MSCD as indicated in table I.

6. MSCD at 1,000 hours shall be not less than 80 percent of initial MSCD when measured at rated voltage.

7. Coating adhesion: Applicable for exterior coated colored lamps.

8. High temperature: Applicable for qualification tests only (185 °F [85 °C]).

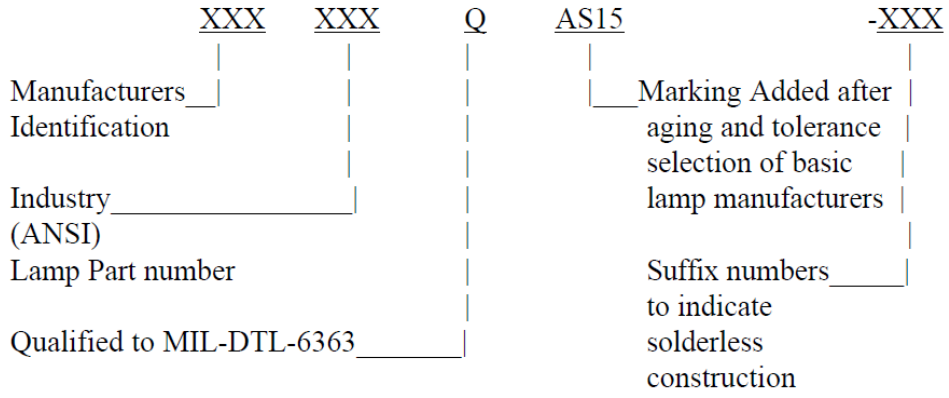
9. Thermal shock: Applicable for qualification tests only. Use temperatures for type I lamps.

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10. Random vibration: Applicable for qualification tests only. See figure 2 for random vibration curve.
11. Shock: Applicable for qualification tests only. See table I for applicable "g" level.
12. Humidity: Applicable for qualification tests only.
13. Salt spray: Applicable for qualification tests only.
14. Marking: Production lamps qualified to MIL-DTL-6363 and listed on QPL No. 6363 shall be marked in accordance with the preferred marking below. as . The letter "Q" shall be incorporated into the method of designation.

EXAMPLE

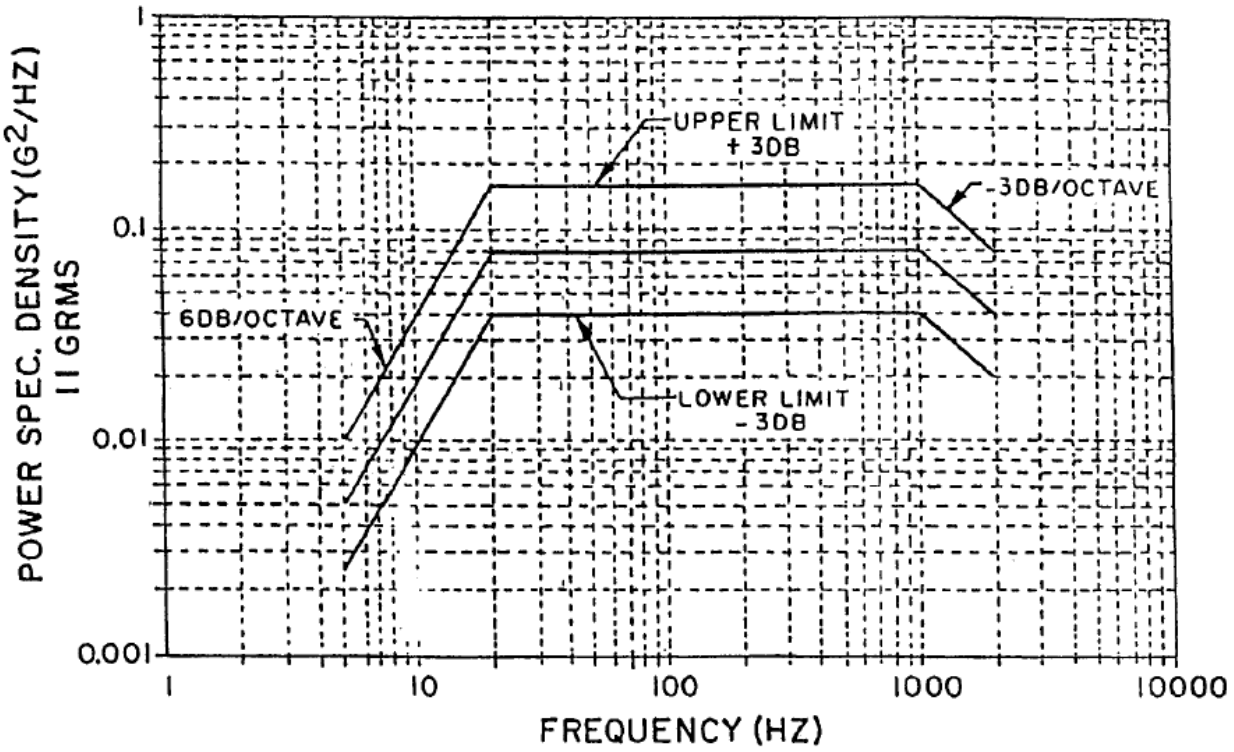
Preferred Marking



Alternate Acceptable Marking

Where marking area is limited by base size, length, or size of marking, or where pre-marked bases are used; the letter "Q" may be marked on any convenient space on the base.

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Freq Hz	Vibration Level PSD (G^2/Hz)	Upper Limit (+3 DB) PSD (G^2/Hz)	Lower Limit (-3 DB) PSD (G^2/Hz)
5	0.005	0.01	0.0025
10	0.02	0.04	0.01
20	0.08	0.16	0.04
1000	0.08	0.16	0.04
2000	0.04	0.08	0.02

FIGURE 2. Random vibration curve for T-1-3/4 lamps.

15. Interchangeability: These lamps may be similar to, but are not interchangeable with, other commercial or military lamps due to unique aircraft applications and qualification.

16. Changes from previous issue: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
- Army – AV
- Navy – SH
- Air Force – 99

Review activities:
- Navy – AS, MC
- Air Force – 71
- DLA – GS

Preparing activity:
- Navy – SH
- (Project 6240-2023-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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